Unit-Step versus Batched Plastic Reduction

Exact Non-Equivalence and Lattice Equivalence

Technical note

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Abstract

The unit-step and nearest-integer-shear versions of the two-dimensional plastic-reduction algorithm are *not* identical algorithms: even in exact rational arithmetic, they can terminate at different matrices in the elastic domain. The difference is therefore not caused by floating-point error. What is true is an exact lattice-equivalence statement. Every output of either algorithm is related to the input, and hence to the other output, by an integer unimodular congruence. After a consistent canonicalization to one fundamental wedge, generic interior outputs agree. This note proves these statements and gives a counterexample whose initial point has Poincare radius approximately 0.636.

1 Setting and the two algorithms

Write a symmetric positive-definite metric as

$$C = \begin{pmatrix} a & c \\ c & b \end{pmatrix}, \qquad a > 0, \quad b > 0, \quad ab - c^2 > 0,$$

and define the elastic domain

$$\mathcal{E} = \left\{ C \succ 0 : \ |c| \le \frac{1}{2} \min(a, b) \right\}.$$

The upper and lower integer shears are

$$U_m = \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix}, \qquad V_m = \begin{pmatrix} 1 & 0 \\ m & 1 \end{pmatrix}, \qquad m \in \mathbb{Z}.$$

They act by congruence, $C \mapsto W^\mathsf{T} C W$. Explicitly,

$$U_m^\mathsf{T} C U_m = \begin{pmatrix} a & c + ma \\ c + ma & b + 2mc + m^2 a \end{pmatrix}, \tag{1}$$

$$V_m^\mathsf{T} C V_m = \begin{pmatrix} a + 2mc + m^2 b & c + mb \\ c + mb & b \end{pmatrix}. \tag{2}$$

Outside $\mathcal{E}$, the current unit-step implementation chooses

$$\begin{cases} U_{-\operatorname{sgn}(c)}, & a \le b, \\ V_{-\operatorname{sgn}(c)}, & b < a, \end{cases}$$

then tests the branch and stopping conditions again. The batched candidate instead uses the nearest integer

$$m = \begin{cases} \operatorname{nint}(-c/a), & a \le b, \\ \operatorname{nint}(-c/b), & b < a. \end{cases}$$

The sampled test used round-half-away-from-zero, but the counterexample below does not lie on a half-integer tie.

2 What can and cannot be batched

For $s \in \{-1, 1\}$ and $k \in \mathbb{N}$,

$$U_s^k = U_{ks}, \qquad V_s^k = V_{ks}.$$

Consequently,

$$(U_s^k)^\mathsf{T} C U_s^k = U_{ks}^\mathsf{T} C U_{ks},$$

and similarly for V. Thus a batched shear is algebraically identical to a fixed run of unit shears in the *same direction*.

This identity does not prove equivalence of the adaptive algorithms. The unit-step method retests both $a \le b$ and membership in $\mathcal{E}$ after every step. During the putative run it may change from U to V, or stop. A nearest-integer shear commits to the entire translation before making either decision. Therefore it can cross a decision boundary that the unit algorithm would have detected.

3 Exact counterexample away from the disk boundary

Consider the rational metric

$$C_0 = \begin{pmatrix} 1 & -151/100 \\ -151/100 & 3 \end{pmatrix}, \qquad \det C_0 = \frac{7199}{10000} > 0.$$

The initial nearest-integer quotient is 151/100, so the batched choice is unambiguously $m = 2$.

Unit-step path

The unit method first applies U_1:

$$C_1^{(u)} = U_1^\mathsf{T} C_0 U_1 = \begin{pmatrix} 1 & -51/100 \\ -51/100 & 49/50 \end{pmatrix}.$$

Now $b < a$ and $51/100 > (49/50)/2 = 49/100$, so it does not stop; it applies V_1 and obtains

$$C_u = V_1^\mathsf{T} C_1^{(u)} V_1 = \begin{pmatrix} 24/25 & 47/100 \\ 47/100 & 49/50 \end{pmatrix}.$$

This endpoint lies strictly inside $\mathcal{E}$ because $47/100 < \frac{1}{2}(24/25) = 48/100$.

Batched path

The batched method first applies U_2:

$$C_1^{(b)} = U_2^\mathsf{T} C_0 U_2 = \begin{pmatrix} 1 & 49/100 \\ 49/100 & 24/25 \end{pmatrix}.$$

It then applies V_{-1} and obtains

$$C_b = V_{-1}^\mathsf{T} C_1^{(b)} V_{-1} = \begin{pmatrix} 49/50 & -47/100 \\ -47/100 & 24/25 \end{pmatrix}.$$

This endpoint is also strictly inside $\mathcal{E}$. In particular,

$$C_u \neq C_b$$

in exact rational arithmetic. The difference is order one and cannot be attributed to floating-point roundoff.

The Poincare coordinates used by the plotting code are

$$x_p = \frac{a-b}{a+b+2\sqrt{\det C}}, \qquad y_p = \frac{2c}{a+b+2\sqrt{\det C}}.$$

For C_0 this gives

$$(x_p, y_p) = \left(-\frac{100}{200+\sqrt{7199}}, -\frac{151}{200+\sqrt{7199}}\right) \approx (-0.3511, -0.5301),$$

with radius $r_p \approx 0.6358$. Hence the phenomenon occurs well inside the disk, not only near its numerically ill-conditioned boundary.

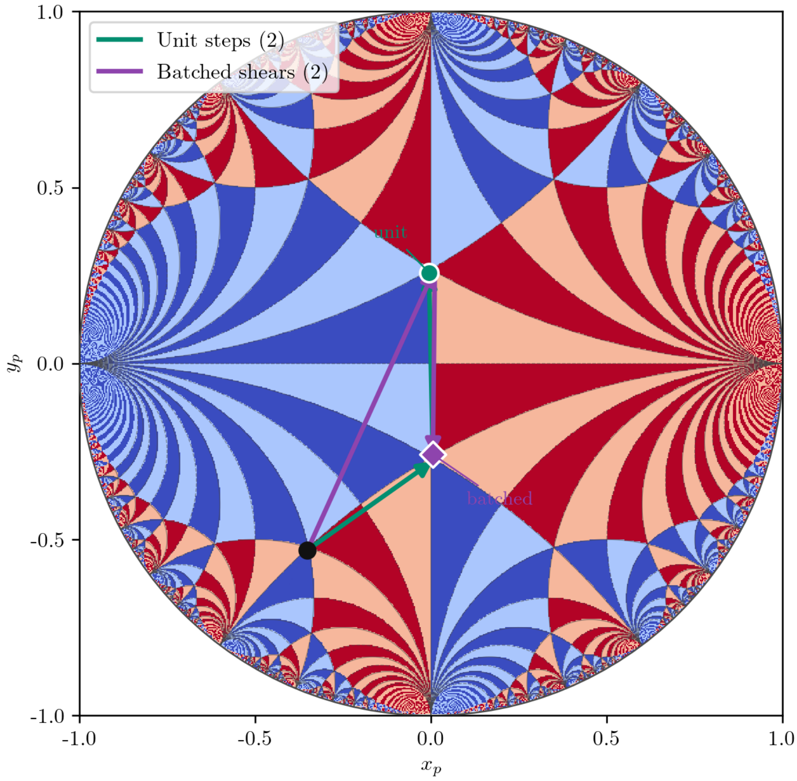

Figure 1: The exact rational counterexample. The green unit-step path and the purple batched path start at the same interior point and terminate at different representatives of the elastic domain.

4 The equivalence that is exact

4.1 Unimodular-orbit equivalence

Proposition. Assume both algorithms terminate. Let $M_u, M_b \in \mathrm{SL}(2,\mathbb{Z})$ be their accumulated right-acting transformations, so that

$$C_u = M_u^\top C_0 M_u, \qquad C_b = M_b^\top C_0 M_b.$$

Then C_u and C_b are exactly congruent under $\mathrm{SL}(2,\mathbb{Z})$.

Proof. Set $Q = M_u^{-1} M_b \in \mathrm{SL}(2,\mathbb{Z})$. Then

$$Q^{\mathsf{T}} C_u Q = Q^{\mathsf{T}} M_u^{\mathsf{T}} C_0 M_u Q = (M_u Q)^{\mathsf{T}} C_0 (M_u Q) = M_b^{\mathsf{T}} C_0 M_b = C_b. \quad \square$$

For the counterexample,

$$M_u = U_1 V_1 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad M_b = U_2 V_{-1} = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix},$$

and

$$Q = M_u^{-1} M_b = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Indeed,

$$C_b = Q^{\mathsf{T}} C_u Q.$$

Thus the endpoints represent the same lattice state but are not the same matrix representative.

4.2 Energy and canonical representatives

If the energy satisfies the material symmetry

$$\Phi(Q^{\mathsf{T}} C Q) = \Phi(C) \qquad (Q \in \mathrm{GL}(2,\mathbb{Z})),$$

then the two endpoints have exactly the same energy in exact arithmetic. This is the physically relevant equivalence for a $\mathrm{GL}(2,\mathbb{Z})$-invariant model. Tensor components, path histories, and the accumulated transformation matrices need not be identical.

To demand a unique matrix output, one must additionally canonicalize the elastic endpoint to a chosen half-open fundamental wedge, for example

$$\mathcal{F} = \{C \succ 0 : \ a \le b, \quad 0 \le c \le a/2\},$$

with explicit boundary tie rules. The map

$$\tau(C) = \frac{c + i\sqrt{\det C}}{a}$$

sends $\mathcal{F}$ to half of the standard modular fundamental domain: $|\tau| \ge 1$ and $0 \le \operatorname{Re}\tau \le 1/2$. The standard interior uniqueness of the modular fundamental domain then implies that two lattice-equivalent, strictly reduced endpoints have the same canonical representative. Boundary points still require a prescribed half-open convention.

For the counterexample, sign normalization followed by diagonal ordering maps C_b exactly to C_u. This explains why the two paths can be physically equivalent even though the raw matrices disagree.

5 Numerical evidence and conclusion

On the 300×300 Cartesian grid in the Poincare disk, there were 70 168 strictly interior samples. Comparing the raw endpoints, 62 580 agreed within relative and absolute tolerances of 10^{-12}, while 7 588 did not. A subsequent symmetry classification mapped all sampled endpoint pairs to one another by diagonal exchange and/or off-diagonal sign change to numerical precision. These observations agree with the exact analysis: the raw algorithms are not exactly equivalent; their outputs remain in the same integer-unimodular orbit; a canonicalization step is needed if downstream code requires the same matrix representative; and large-radius inputs can amplify floating-point error, but roundoff is not the cause of the representative change.

Therefore a nearest-integer batched implementation is safe only when the application compares states modulo lattice symmetry, or when it canonicalizes the final result. It is not a drop-in replacement if the exact raw endpoint, the path, or the accumulated transformation is part of the required output.